

External noise reduction using WhisperMask, a mask-type wearable microphone

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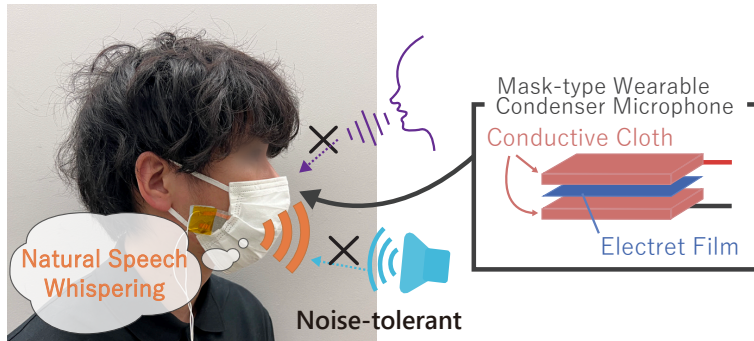
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Use Case of WhisperMask



Figure 1: WhisperMask, a wearable electret condenser microphone that can only catch the user's voice even in noisy environments. The microphone is shaped like a mask, making it hands-free, wearable, and unobtrusive, and lightweight device.

ABSTRACT

Conversation through voice is one of the basic means of human communication. Online communication systems and voice user interfaces for operating smartphones or smart devices are increasingly important as an interface that anyone can use daily. However, voice input is difficult to use in noisy environments or in environments where multiple people speak at the same time because unintended voices are included. Conventional microphones such as pin microphones or contact-type microphones such as throat microphones, and NAM microphones have been proposed for reducing external noise, however, pin microphones are not tolerant of high volume, and contact-type microphones generate contact noise when the user nods or walks. We propose WhisperMask, a

wearable electret condenser microphone that can acquire speech even in noisy environments. In addition, the proposed mask-shaped microphone allows hands-free input and is wearable, unobtrusive, and lightweight interface.

CCS CONCEPTS

• **Human-centered computing** → *Mobile devices*; **Sound-based input / output**.

KEYWORDS

microphone, speech, wearable device, mask-type interface, speech enhancement, noise cancelling

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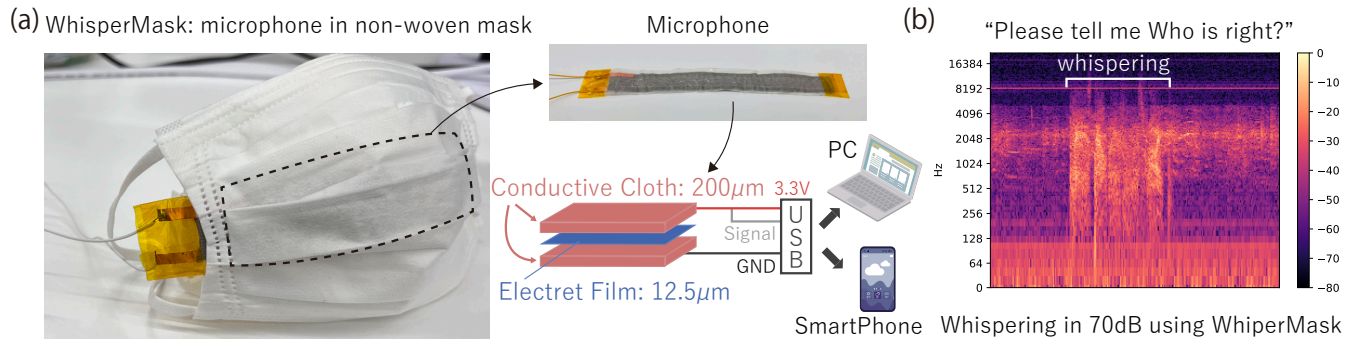


Figure 2: (a) A prototype of whispermask, mask-type microphone woven into a non-woven mask. (b) The electret condenser microphone made with film and silver plated fabric. The recorded sound was sent to a PC or smartphone via USB (c) An example of recorded whispering voice "Please tell me who is right?" [3] using WhisperMask, where the volume of ambient noise is 70dB

1 INTRODUCTION

Conversation through voice is one of the basic means of human communication. Voice user interface (VUI) is a method of interactive operation with devices through speech recognition, and has been widely used in smartphones, smart devices, and car navigation systems as an interface that anyone can use on a daily basis. Also, due to COVID-19, online communication has been increasing important and a system that allows voice communication without location restrictions is required. However, voice input is difficult to use in noisy places or in environments where multiple people speak at the same time, because unintended voices are included.

A pin microphone has been used as a method to catch only one's own voice. This method increases directivity by restricting the direction of sound input to the microphone or by using multiple microphones. However, it only limits the direction from which the incoming sound comes so that it is not tolerant of loud noise.

In addition, throat microphones and Non-Audible Murmur microphones (NAM microphones) [14], which convert vibrations transmitted over the surface of the body into sound, have been proposed as a method to pick up only the user's own voice. These devices do not catch external noise coming through the air and thus can be used in noisy locations, however, they require the user to maintain contact with the device, because the user's natural neck movements, such as nodding, turning the head, or walking around, generate noise.

This study proposes a wearable microphone, WhisperMask, which is a wearable mask-type microphone that is an extension of the electret condenser microphone and can acquire speech even in noisy environments. The WhisperMask is an extension of the electret condenser microphone in the form of a wearable mask that can capture speech even in noisy environments. In addition, the proposed mask-shaped microphone allows hands-free input and is wearable, unobtrusive, and lightweight interface.

Experimental results showed Signal to Noise Ratio (SNR) of WhisperMask is better than that of pin microphone even in noisy environment. Also, speech recognition rate using WhisperMask exceeded

the recognition rate with a pin microphone using noise reduction software [1].

2 RELATED WORK

2.1 Wearable Microphone

With the miniaturization of semiconductors and advances in Micro Electro Mechanical Systems (MEMS) technology, high-precision microphones have become readily available. Here, we clarify the position of the WhisperMask proposed in this research compared to existing microphones.

2.1.1 Contact type microphone. Throat microphones [5, 16], NAM microphones [10, 14], and bone-conduction microphones exist as contact-type microphones. These microphones can acquire vibrations transmitted to the surface of the body or bones by providing a medium such as silicon between the diaphragm of the microphone and the body. Since these microphones transmit sound directly from the body to the microphone without passing through the air, they can prevent the introduction of sounds transmitted through the air, such as external sounds, thereby reducing noise. On the other hand, these microphones require contact, which means that body movement becomes a natural source of noise, that weak components such as whispers and certain frequency bands are cut off in the process of passing through body tissues, and that they require post-processing to convert the sound into audible speech.

2.1.2 Directional Microphone. The most common wearable but non-contact microphones are pin microphones and headset microphones. These are used to get the user's voice by putting it on clothing or placing it near the mouth. Some are unidirectional microphones, which physically limit the direction of sound input. Unlike contact-type microphones, these microphones do not require post-processing for conversion and can acquire small sounds such as whispers, but on the other hand, they have low tolerance to loud noise.

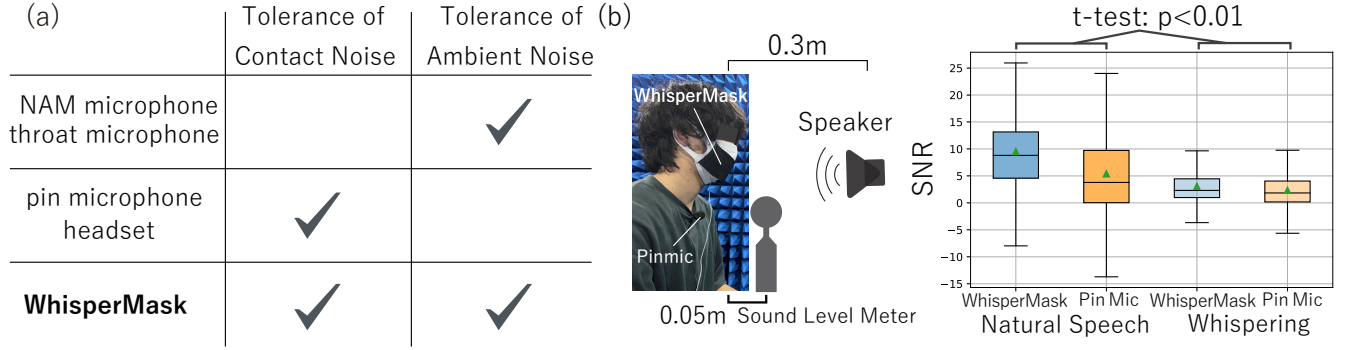


Figure 3: (a): WhisperMask’s position in wearable microphones. (b)View of the environment for measuring voice volume, recorded with WhisperMask and pin microphone(PinMic) using a precision sound level meter located 0.3m away. Measurements of voice volume in two utterances: Natural Speech, and Whispering

2.2 Software-based Speech Enhancement

In the field of speech research, in order to distinguish user’s voice from other noise, speech enhancement has been studied[4]. In conventional acoustic methods using adaptive system to background noise[6] or e spectral amplitude[2] was proposed. With the development of deep learning model, speech enhancement methods by software are proposed[9]. For generating a noise reduced audio accurately, speech enhancement using generative models are proposed[11, 12]. In especially, high accurate and real-time speech enhancement using CNN[15, 17], and U-Net[1] also has been proposed.

3 WHISPERMASK: WEARABLE, NOISE-SUPPRESSIVE MICROPHONE

Wearable microphones for daily use such as pin microphone, headset are not tolerant of loud noise, and conventional contact-type microphones such as throat microphones, NAM microphones generate noise when the users perform natural movements such as nodding and turning their head, and walking, which they do on a daily basis. Also, contact-type microphone catch the vibration from the surface of their skin, and the recorded sound is filtered by the body, therefore acoustic characteristics are different and specialized speech recognizers or speech enhancements are needed.

To solve these problems, we propose WhisperMask, a wearable electret condenser microphone that can acquire the user’s voice even in noisy environments. Even in the noisy places, not only natural speech, WhisperMask can get low volume voice like whispering, which get lost in noisy environments (Fig. 2(b)).

Also, WhisperMask is shaped like a mask, making it hands-free, wearable, and unobtrusive. In addition, the user’s voice can be emphasized without software tuning, so this mask-type microphone can be used with existing devices without modification such as PCs, smartphones, and smart devices.

Finally, the microphone is not in direct contact with the skin and is less affected by facial and body movements than throat or NAM microphones. It is further superior in that it does not generate pop noise that occurs when one blows on it. Finally, the microphone is in the shape of a mask, making it hands-free and inconspicuous.

It is lightweight at 30 g, including the mask, which reduces the burden on the user.

3.1 Device Configuration

Our microphone consists of an electret capacitor made of a plastic film and conductive cloths (Fig.2(a)). The film was consisted of PFA (Per Fluoro Alkoxy polymer) which is well-known to be easily charged and performs as a electret material. We used a PFA film with a thickness of 12.5 μm . Adhesive tape was applied around the perimeter of the film for wind protection, and silver plated polyester cloths with a thickness of 200 μm was fixed on both sides. This formed the structure of an electret capacitor with the electret sandwiched between two electrodes. The size of the condenser is designed to be 18 cm x 7 cm as shown in Fig. 2(a).

The principle of operation as a microphone is the same as that of a general electret condenser microphone. When the electret is subjected to voice vibration, the distance between the electret film and the electrode changes with the vibration, and the potential on the electrode changes. This allows the sound to be extracted as a voltage amplitude.

To use the microphone in the experiment, the microphone part of a commercially available condenser microphone (FIFINE K053) was disconnected from the circuit, and two conductors connected to the circuit were soldered to two pieces of silver plated polyester, which are the electrodes of the electret condenser. The two conductors were soldered to the two pieces of silver plated polyester fabric that would serve as the electrodes of the electret condenser. Since the fabric cannot withstand the temperature of direct soldering, the soldering part was reinforced with copper tape. The USB port attached to the circuit was plugged into a PC and Smartphone recognized it as a normal microphone device and collected sound.

3.2 Experiment1: SNR of WhisperMask and pin microphone in 65dB background noise

We measured the SNR of WhiperMask and pin microphone with English phrases while the loudness of the ambient noise was kept constant to 65dB. The pin microphone used was commercially available, unidirectional microphone (FIFINE K053). Ambient noise,

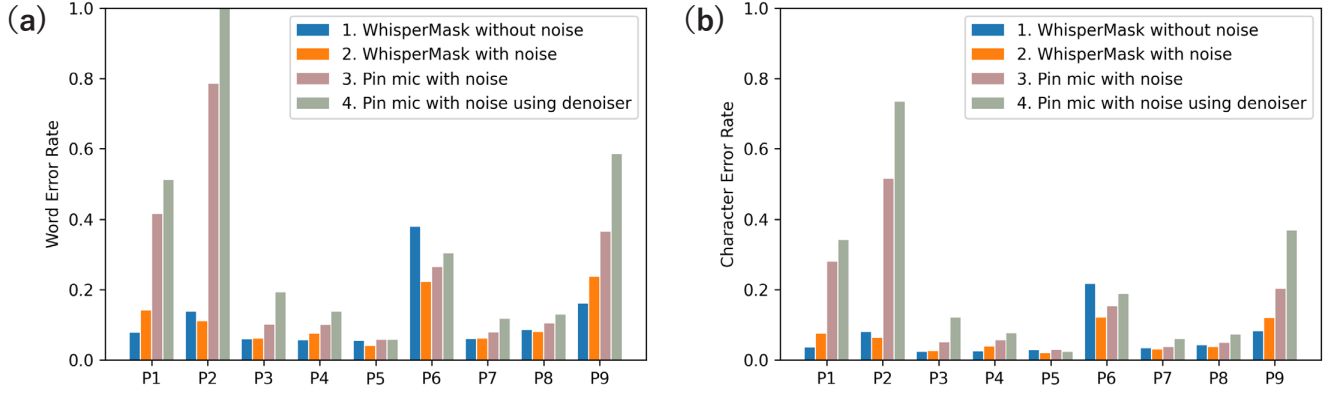


Figure 4: The result of comparison between WhisperMask in noisy environment and Pin Mic software denoiser[1] (a): Word Error Rate(WER) metrics, (b): Character Error Rate(CER) metrics

which is also used in CSTR corpus, was played through an omnidirectional speaker from a distance of 0.3 m. The background noise was set to 65 dB at the mouth of the speaker. The CSTR corpus is large dataset for NAM recognition. The speaker wore two microphones, a WhisperMask and a pin microphone on the speaker's chest, and the two microphones simultaneously recorded the speaker's voice as the speaker reads phrases. The WhisperMask and the pin microphone were attached and detached each time the speaking style was changed: Natural Speech and Whispering. There were 10 speakers reading, 4 males and 6 females with an average age of 24.6 years. The 100 phrases to be read out were selected from the MacKenzie and Soukoreff dataset[7], which was a large corpus for text entry and easy to understand for non-native English users.

In order to measure SNR, We used Monterial Force Alignment[8] for detecting speech segments in the audio clips. The pretrained English (US) ARPA models and dictionaries were used to detect speech segments.

Fig. 3(b) summarizes the SNR of all speakers' utterances by speaking style: Natural Speech and Whispering. The t-test between WhisperMask and pin microphone is $p=2.38e-51(<0.01)$ for Natural Speech and $6.45e-10(<0.01)$ for Whispering, indicating that WhisperMask is statistically higher SNR than pin microphone.

3.3 Experiment2: Software Speech Enhancement vs Hardware Noise Cancelling

In the field of speech research, speech enhancement for reducing background noise has been proposed[4, 9]. Recently, real-time speech enhancement is so high accurate that we evaluate our microphone compared with software-based noise cancelling method using deep learning[1].

As a data collection, We collected 9 subjects and they respectively pronounced 100 sentences selected as mentioned above. Then, we collected 4 types of audio data. (c-1): Natural speech data recorded with WhisperMask, (c-2): Natural speech data recorded with WhisperMask in noisy environment, (c-3): Natural speech data recorded with a pin microphone in noisy environment, (c-4): Natural speech data recorded with a pin microphone in noisy environment, which

was enhanced using denoiser[1]. The background noise is the same as *Experiment1*.

With the metric of Word Error Rate(WER) and Character Error Rate(CER), We evaluate the accuracy of speech recognition using OpenAI Whisper[13], a transformer based speech recognizer. As the result, WER and CER of WhisperMask in noisy environment(c-2) are the same as that of Natural speech in not noisy environment(c-1) and more than that of pin microphone(c-3) and pin microphone with denoiser(c-4) as shown in Fig. 4.

4 DISCUSSION

In the Case2, WER and CER of pin microphone using denoiser(c-4) tend to be less than that of original pin microphone (c-3). This could be because the data converted by denoiser drops together the acoustic features needed for speech recognition. In addition, the P2's WER of pin microphone using denoiser (c-4) was over 1.0, this is because the speech recognizer repeatedly output the misrecognized results when the recognizer failed.

5 CONCLUSION

We propose WhisperMask, a mask-type wearable electret condenser microphone that can catch the user's speech clearly even in noisy environments compared to a pin microphone in terms of SNR and WER/CER. WhisperMask is also hands-free, lightweight, and unobtrusive interface, so WhisperMask provides the user to communicate with people or devices through speech modality without location restrictions.

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